A Reproducible Lunar Preprocessing Pipeline for Vision Foundation Model Evaluation

Alexander Kerr[1,2], Mark L. Carroll[1], Jordan A. Caraballo-Vega[1], Melanie J. Frost[1,2], Roger L. Gill[1,2], <others>

[1]Data Science Group, NASA Goddard Space Flight Center; [2]ADNET Systems, Inc

Motivation

- **Lunar foundation model for science**

Foundation models are a useful tool for quick iteration, as finetuning can be done on a small amount of samples.

- **Scalable and efficient workflow**

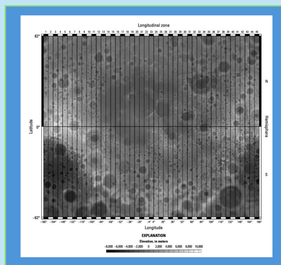
Workflow needed to be configurable to allow for ablations, but also streamlined to allow users to quickly begin experimenting.

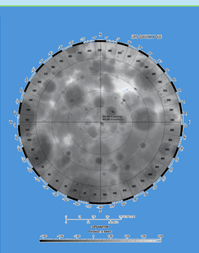




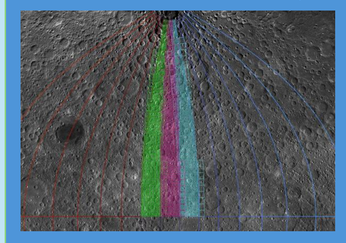
Artemis missions provide a motivation for expanded Lunar science work (above, left); Lunar Reconnaissance Orbit (LRO) provided a large corpus of data to study from (above, right)

Dataset Creation





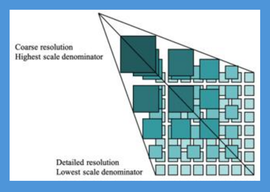
LTM and LPS zones (above)



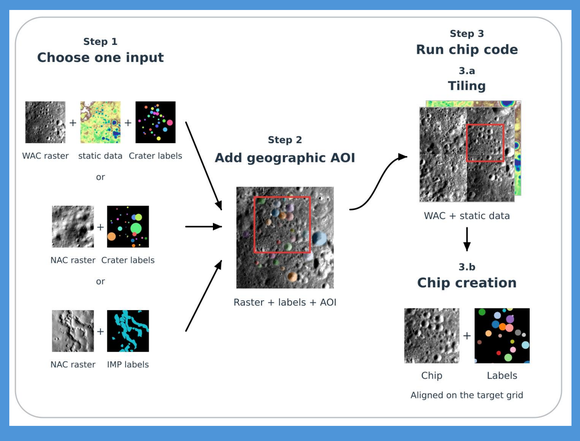
LTM zones on Lunar base image (left), zone level subdivision example (right)

Armstrong Tiling Scheme

- Aids in ML dataset creation by dividing Lunar surface evenly for processing
- Separates the moon into 90 Transverse Mercator, 2 Polar zones
- TM and Polar zones are equally sized/spaced
- Zones are subdivided into grids at a given zoom level; 15/26 zoom levels per Polar/TM zone

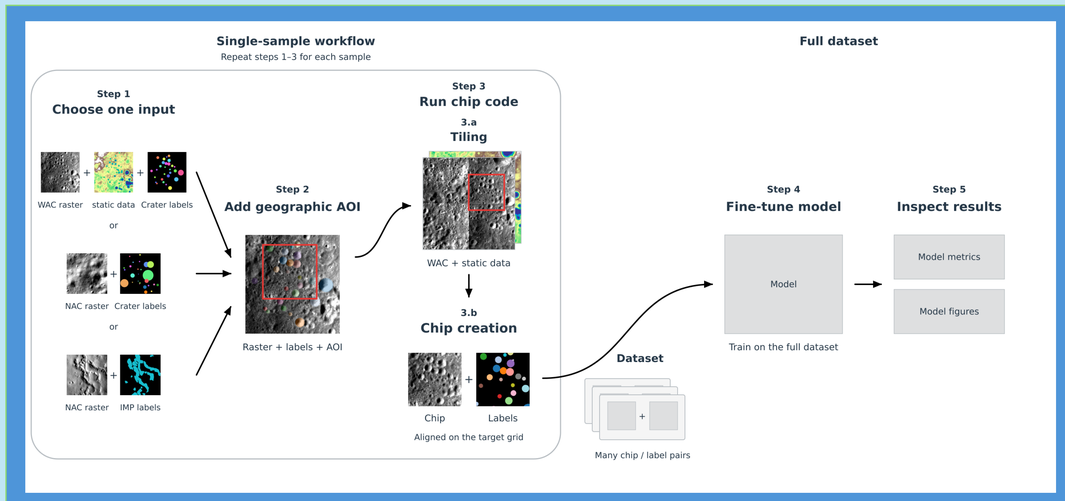


Chip Creation Workflow



- Allows users to create high-quality datasets from labeled data quickly
- Custom tool created to develop crater labels manually, incorporating Robbins catalog
- Custom tool created to process input data with labels and geographic AOI into chips
- Multiple data modalities: WAC (7-band, Vis + UV), NAC (single-band), Static (68 bands of various types)
- Multiple ML tasks: crater detection (semantic/instance segmentation), IMP detection (semantic segmentation)

Finetuning FMs



Performance of dinov3 vs lfm on varying band counts

Results

References

[1] Neigh C.S.R, et all (2018) Enhanced Very-High Resolution (EVHR) Products for NASA's Earth Science Investigators. Oral presentation conducted at the Earth Science Technology Forum (ESTF) in Silver Spring, MD.
[2] Caraballo-Vega, J. A., et al. "Optimizing WorldView-2,-3 cloud masking using machine learning approaches." Remote Sensing of Environment 284 (2023): 113332.
[3] Thomas, Nathan, et al. "Fusion approach for remotely-sensed mapping of agriculture (FARMA): A scalable open source method for land cover monitoring using data fusion." Remote Sensing 12.20 (2020): 3459.

Acknowledgements

Thanks to the NASA Center for Climate Simulation for providing the computational resources. This work is the result of a collaborative effort between several members of NASA's Land Cover Land Use Change program and the Computational & Information Sciences and Technology Office (CISTO) Data Science Group. DigitalGlobe/Maxar data were provided by NASA's Commercial Archive Data for NASA investigators (http://cad4nasa.gsfc.nasa.gov) under the National Geospatial-Intelligence Agency's NextView license agreement. We acknowledge all the people that contributed and supported this work. This work was supported by NASA's Land Cover Land Use Change program [grant numbers NNH16ZDA001N-LCLUC, NNH20ZDA001N-LCLUC].